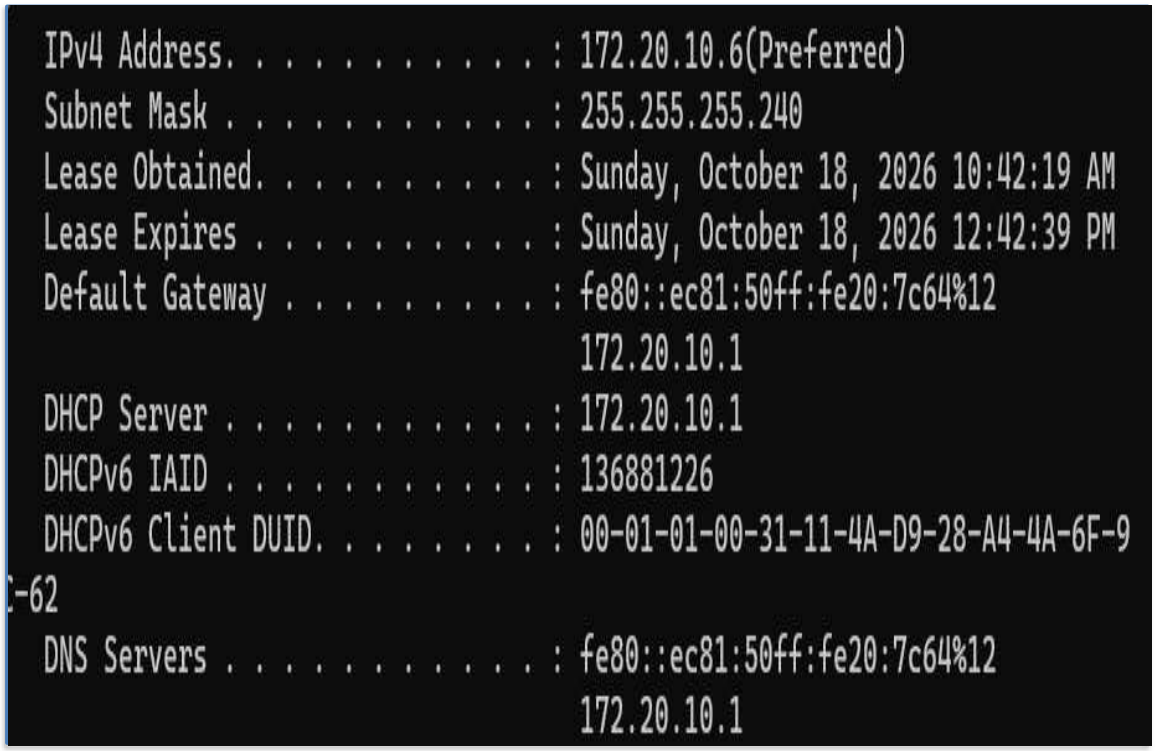
DATA COMMUNICATION & NETWORKING

Week 1 Hands-On Lab Submission Template

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Course: Data Communication & Networking

Slide 2 — Network Configuration

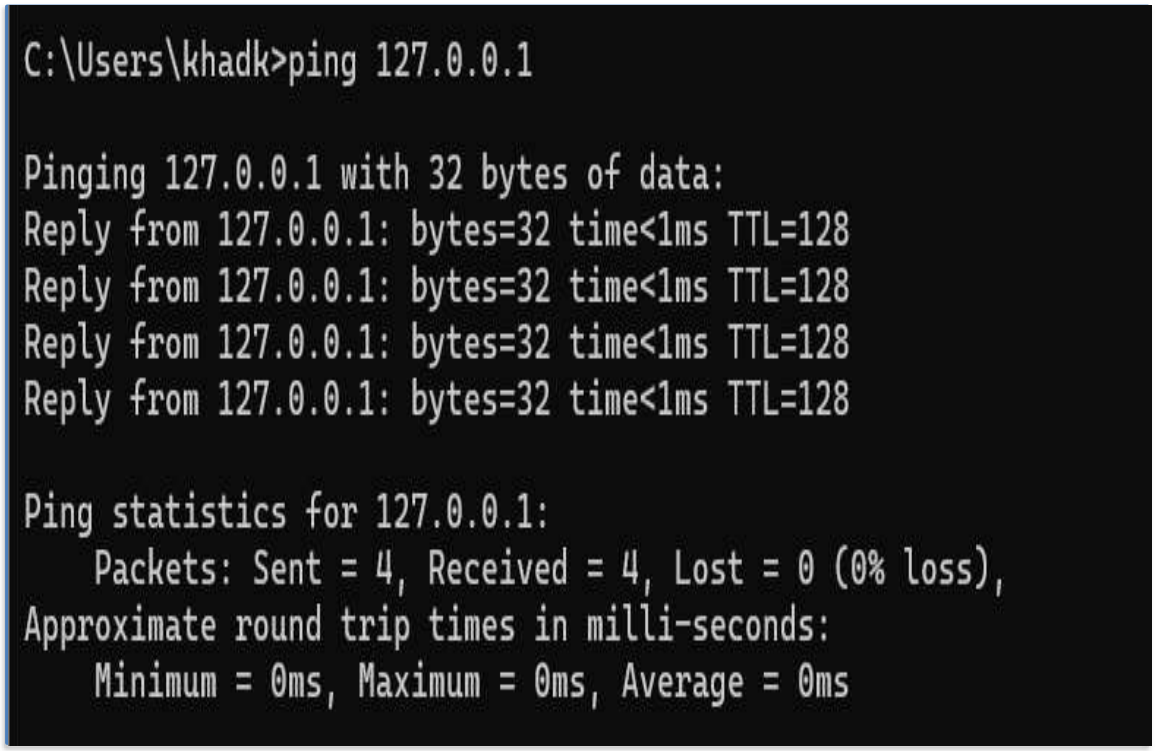
Insert your ipconfig /all screenshot. Type the adapter type, IPv4 address, subnet mask, default gateway, and DNS server(s) in the response area.



Adapter type : Wi-Fi
IPv4 address : 172.20.10.6
Subnet Mask : 255.255.255.240
Default gateway :
fe80::ec81::50ff::fe20:7c64%12
DNS servers:
fe80::ec81::50ff::fe20:7c64%12
172.20.10.6

Slide 3 — Local TCP/IP Test

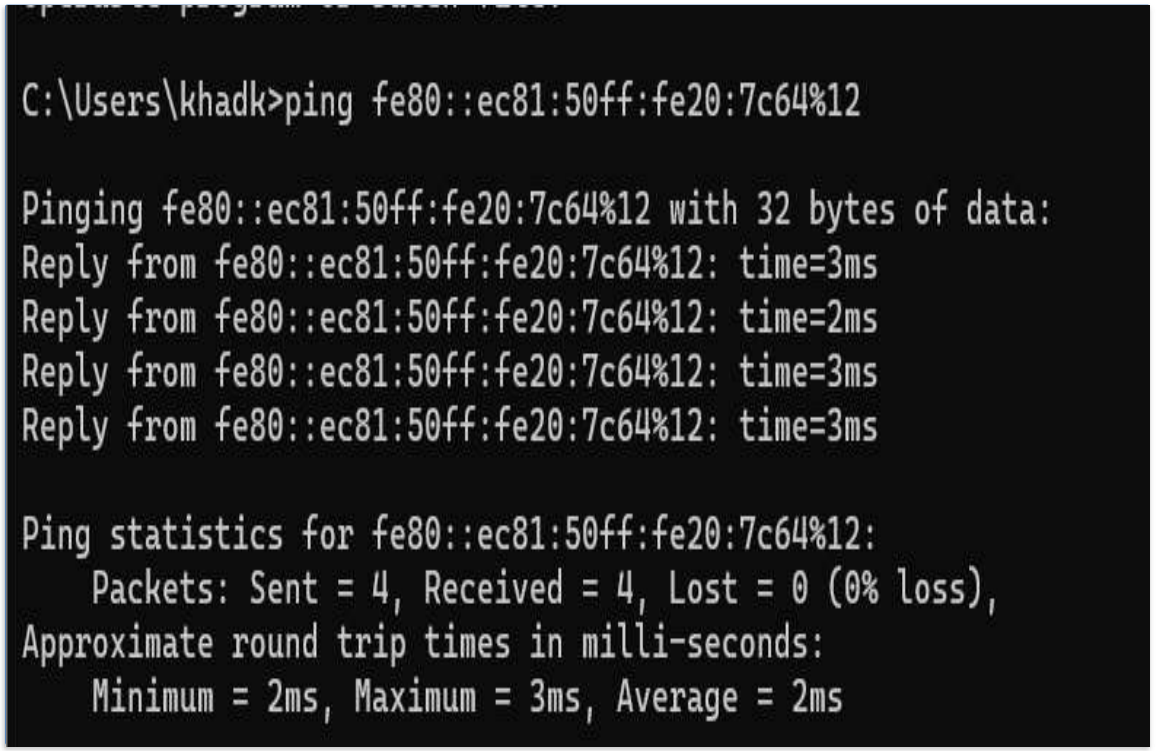
Insert your ping 127.0.0.1 screenshot and briefly explain what the test verifies.



The ping 127.0.0.1 test verifies that my computer's TCP/IP networking software is working correctly. The successful replies show that the local network stack is functioning properly.

Slide 4 — Default Gateway Test

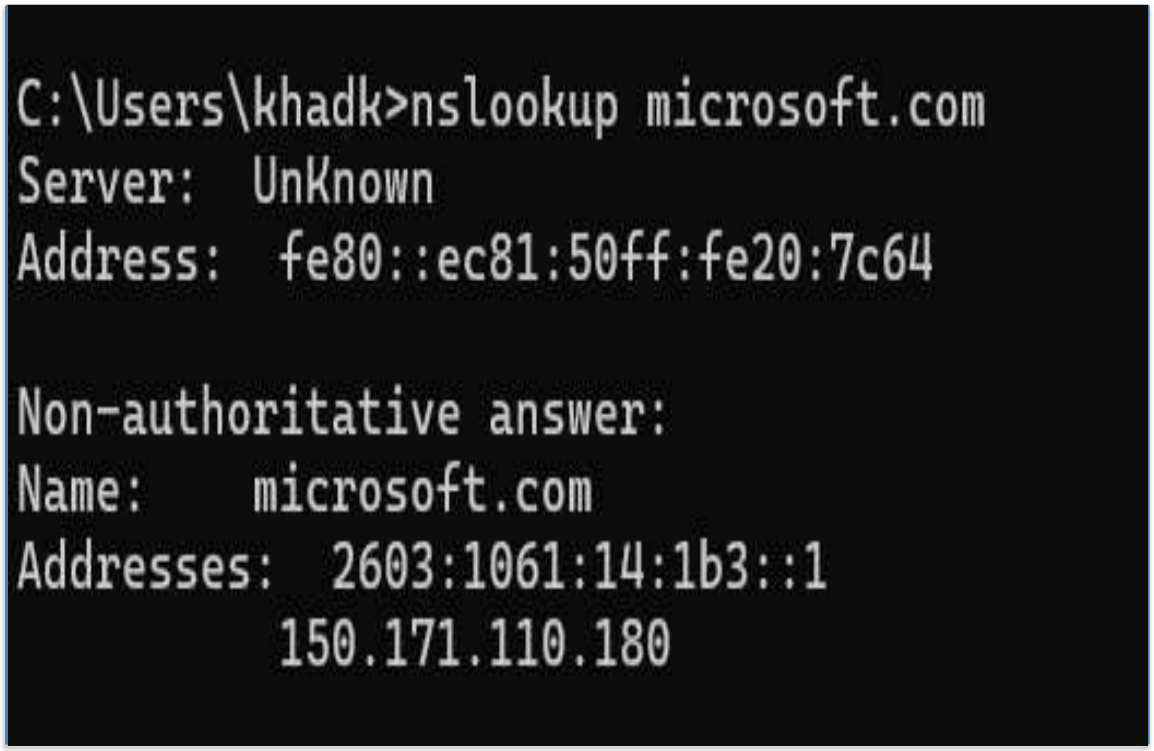
Insert the default-gateway ping screenshot. State whether it succeeded and what the result means.



Yes, the default gateway responded successfully to the ping request with 0% packet loss. This result means that my computer can communicate with the router on the local network.

Slide 5 — DNS Test

Insert the nslookup microsoft.com screenshot. Identify the DNS server and at least one returned IP address.



The DNS server is fe80::ec81:50ff:fe20:7c64. One returned IP address for microsoft.com is 150.171.110.180.

TYPE REQUIRED VALUES / EXPLANATION HERE

Slide 6 — Route Test

Insert the tracert microsoft.com screenshot and briefly explain what a hop represents.

INSERT SCREENSHOT HERE

TYPE REQUIRED VALUES / EXPLANATION HERE

Slide 7 — Analysis Questions 1–4

Answer Questions 1–4 from the lab instructions in complete sentences.

- I am using a Wi Fi adapter. It allows my computer to connect to a wireless network and access other devices and the internet without using a cable.
- My IPv4 address is 172.20.10.6. Yes, it is a private IP address because it falls within the private address range 172.16.0.0 to 172.31.255.255.
- A default gateway is a router that connects a local network to other networks, including the internet. A computer needs a default gateway to send data to destinations outside its local network.
- A DNS server translates domain names into IP addresses. It is useful because users can type website names such as google.com instead of remembering numerical IP addresses.

Slide 8 — Analysis & Reflection

Answer Questions 5–8 from the lab instructions. Your reflection must be in your own words.

- The command ping 127.0.0.1 tests the local TCP/IP network stack on the computer. It does not test the network adapter, router, or internet connection.
- Yes, the gateway responded successfully with 0% packet loss. This proves that my computer can communicate with the default gateway on the local network, but it does not prove that the internet connection is working or that all network services are available.

TYPE YOUR RESPONSE HERE

- This lab helped me understand how network devices such as routers and Wi Fi adapters work together to provide network connectivity. I used IP addressing to identify my device and the default gateway on the network. The ping command showed how the ICMP protocol is used to test communication between devices. I also learned how DNS servers translate domain names into IP addresses, making it easier to access websites and other online resources.